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State of health prediction of lithium-ion batteries based on bidirectional gated recurrent unit and transformer

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ABSTRACT

Lithium-ion batteries have been widely used in various aspects of our lives, playing a crucial role in numerous applications. The state of health (SOH) serves as a pivotal indicator, and accurate prediction of SOH is essential for the safe utilization, management, and maintenance of lithium-ion batteries. In order to accurately predict SOH, a hybrid prediction model by combining bidirectional gated recurrent unit (BiGRU) and Transformer with multi-head attention mechanism (AM) is proposed, which can effectively address the challenge of long time series prediction. In the proposed prediction model, the indirect health indicator (HI), which can characterize the degradation of lithium-ion batteries, is fed into the BiGRU to learn the hidden states of the input features and thus further extract time series features. On this basis, multiple attention is given to the Transformer encoder layer and the input feature vectors, which gives it a better performance in the long-term dependence of the time series. The study based on the lithium-ion battery data from NASA Prediction Center of Excellence (PCoE) shows that the proposed BiGRU-Transformer model has higher accuracy, better robustness and generalisation capability.

1. Introduction

Lithium-ion battery technology is becoming the most suitable device for storing electrical energy in mobile devices and electric vehicles due to its inherent and attractive characteristics: light weight, high energy density, small size, low memory effect, long life and low pollution [1]. However, the capacity decline and power decrease that occur during the use of lithium-ion batteries will inevitably affect their safety performance. Worse still, if lithium-ion batteries experience electrolyte leakage and micro-short circuits, it can lead to battery failure and thermal runaway [2]. In particular, sudden failures of industrial and light mechanical equipment due to battery failure can cause significant economic losses in industrial production. Therefore, it is crucial to quickly and accurately predict the SOH of lithium-ion batteries.

The SOH prediction methods can be divided into three categories: model-based methods, data-driven methods, and hybrid-based methods [3]. Model-based methods have been widely used to predict the battery performance [3,4]. Hosseininasab et al. proposed a SOH estimation method based on a reduced-order electrochemical model to jointly estimate the capacity and resistance with a low dependence on electrochemical parameters [5]. Nejad et al. applied the dual extended Kalman filter algorithm to update the parameters of the second-order equivalent circuit battery model, which allows for the obtaining of accurate estimates of the state of charge in real time [6]. However, the mechanical model brings a huge burden to the battery management system when solving complex partial differential equations [7].

Data-driven prediction methods are mainly based on machine learning (ML) instead of physical models. Data-driven methods use large amounts of historical monitoring data and machine learning technology to build relevant mathematical models or obtain weighting parameters to predict the degradation behavior of lithium batteries. With the development of computer hardware, data-driven methods have become less complex and costly, and easier to apply, and the trade-off between complexity, cost, accuracy and applicability can be made in practice to choose the appropriate method [8]. The typical data-driven methods include relevant vector machines (RVM), support vector regression (SVR) [9], support vector machines (SVM) [10], and neural networks (NNs) [11], etc. For example, Wang et al. proposed a degradation tracking model by fusing health features and Gaussian process regression (GPR) to estimate the SOH of lithium-ion batteries [12]. The

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method based on a dynamical recurrent neural network (RNN) was developed by Chaoui et al. to estimate the SOH [13]. Also based on RNN, You et al. designed a degraded model, which is more practical and suitable for handling continuous data where the battery supports a real-world driving model [14]. Different from RNN, Yang et al. applied a hybrid method based on complete set empirical modal decomposition and SVR to predict the SOH of lithium-ion batteries [15]. Considering that the Wiener process and Ornstein-Uhlenbeck process are capable of modeling long-term degradation trends and short-term volatility respectively, Kong et al. designed a battery health prediction model by fusing the two stochastic processes [16].

On the other hand, the conventional RNNs can effectively solve nonlinear problems using prior information, but the performances of RNNs are limited by the presence of gradient explosion or gradient vanishing during training. Therefore, researchers have created some variants of RNN to solve these problems, such as long short-term memory (LSTM), bidirectional LSTM (BiLSTM), GRU, etc. A hybrid neural network called GRU-CNN was proposed by Fan et al. to estimate the SOH of lithium-ion batteries, which uses the newly observed charging voltage, charging current and charging temperature by learning the shared information and time dependence of the charging [17], while an RNN with GRU was used to estimate the state of charge by Yang et al. in Ref. [18]. Ungurean et al. designed a complete battery management system to optimize NNs by Adam algorithm [19]. The experimental results show that GRU requires significantly fewer parameters although the error of GRU is slightly higher than that of LSTM. Meanwhile, Chung et al. demonstrated experimentally that GRU outperforms LSTM with a small dataset, especially in terms of training epochs [20]. In view of this, Zhang et al. designed an interactive prediction model based on particle filtering-by-time AM-two way gated recursive unit, which is trained offline with historical data [21].

Furthermore, it is worth noting that Transformer architecture has been widely used for sequence modeling due to its ability to capture long-range relationships and high computational efficiency. Transformer is an encoder-decoder architecture that relies entirely on the AM, eschewing CNN and RNN altogether and analyzing the dependencies between inputs and outputs only through AM, and it has been widely used in many fields such as computer vision. Luo et al. identified electrochemical impedance spectroscopy as a new feature and applied Transformer-based NNs to SOH estimation [22]. Chen et al. processed the raw data using a denoising autoencoder and fed the reconstructed sequence into a Transformer network [23]. Experiments were conducted on two datasets and better prediction results were obtained. Hu et al. used the wavelet threshold denoising method to remove the noise signal from the discharge capacity data, and input the processed data into the Transformer model, which was validated using two public sets of lithium-ion battery aging data respectively, to obtain a more accurate prediction result [24]. Gu et al. first applied the Transformer structure to the time series problem and achieved good SOH estimation results [25].

Different from the above two types of SOH prediction methods, hybrid-based methods include a combination of data-driven, filtering techniques, and empirical models. Hybrid methods can effectively exploit the strengths of individual models to obtain accurate SOH. ML methods such as SVM, GPR and NNs are often used in combination with different adaptive modeling techniques to accurately predict SOH. For example, the hybrid model based on GPR and particle filter was used to estimate SOH by Li et al. in Ref. [26].

In summary, data-driven methods based on NNs have good generality and powerful feature extraction capabilities. In particular, RNNs have long been used in sequence modeling tasks to effectively model time series dependencies. Scholars have proposed LSTM/GRU networks to avoid gradient vanishing or exploding in traditional RNNs. However, sequences are still too long for LSTM/GRU networks. Moreover, in most studies, only data from the current time step were imported into the LSTM/GRU network, limiting the ability of the model to study long-term dependencies, resulting in the model focusing more on later inputs than earlier ones, and not fully considering the entire sequence. More importantly, as a variant of RNN, GRU still cannot completely solve the gradient vanishing problem, and has the disadvantage of the RNN structure itself, which has become a key bottleneck for RNN in the trend of gradually increasing data volume and model volume. Fortunately, the Transformer model uses the self-AM to achieve the encoding and representation learning of the input sequences, which has higher parallelism and computational efficiency compared with the traditional RNN or CNN. Therefore, Transformer has obvious advantages in processing long sequences and large-scale data. Besides, Transformer uses a self-attentive mechanism to achieve encoding and representation learning of the input sequence, which can capture the dependencies between different positions in the sequence and achieve context awareness. Therefore, the transformer is good enough to capture long-term dependencies to improve the accuracy and robustness. However, roughly discarding RNNs and CNNs deprives the models of the ability to capture local features.

Therefore, considering the shortcomings of GRU's inability to perform parallel computation, and inspired by Transformer's excellent ability of global feature extraction and BiGRU's ability of sequence feature extraction, we propose a hybrid prediction model, which combines the structures of BiGRU and Transformer and is denoted as BiGRU-Transformer model, to improve the performance of SOH prediction. The proposed model obtains the indirect HI, which can replace the capacity degradation by analyzing the charging and discharging process, extracts the battery time series features using BiGRU, and learns hidden states of the input features. The information got from BiGRU is then passed into Transformer's encoder and decoder to investigate the long-term dependence of capacity degradation in the battery performance data, and the SOH of lithium-ion batteries can be predicted accurately by using multi-head AM to better learn the correlation between the sequence data. Thus, we summarize the main contributions of this paper as follows.

(1) The indirect HI that can reflect the capacity degradation of lithium-ion batteries is extracted and used for the SOH prediction instead of using the capacity directly, and the relationship between the extracted HI and the capacity is present through correlation analysis.
(2) The BiGRU model is used to handle the serial characteristics of the extracted HI for lithium-ion batteries. Compared to the widely used single GRU model, the BiGRU model can better capture the information contained in the context of the sequence.
(3) The BiGRU-Transformer model is proposed by combining the structures of BiGRU and Transformer, which is better able to improve the SOH prediction performance of lithium-ion batteries by learning long-term dependencies.
(4) Three different metrics are used to evaluate the proposed method via the public battery datasets, and the effectiveness and reliability of predicting SOH based on the BiGRU-Transformer model are verified.

The rest of this paper is organized as follows. The basic neural network models used in BiGRU-Transformer model are described in Section 2. Section 3 presents the BiGRU-Transformer model and the corresponding SOH prediction process. In Section 4, the dataset, data preprocessing and experimental parameters are shown, while Section 5 shows the SOH prediction results and comparative analyses. Finally, Section 6 concludes this paper and discusses the possible future directions.

2. Preliminaries

A brief introduction to BiGRU and Transformer used in the BiGRU-Transformer model is presented in this section.

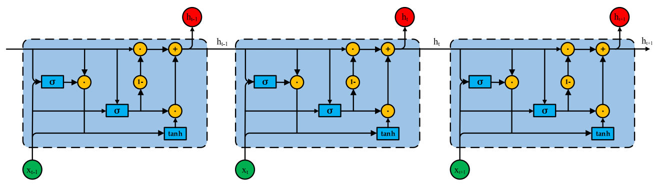

Fig. 1. Recurrent unit structure of GRU.

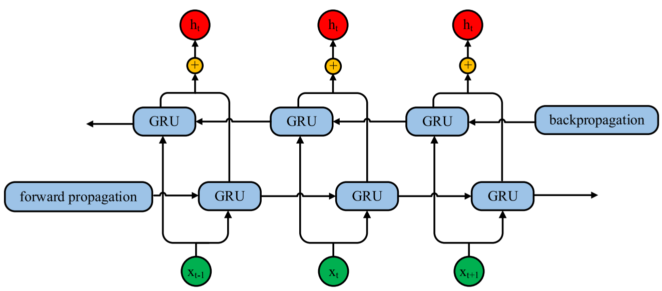

Fig. 2. Recurrent unit structure of BiGRU.

2.1. BiGRU

GRU is a gating mechanism for RNN that overcomes the long-term memory difficulties of traditional RNNs. LSTM has three gates such that the training process to form the network is time-consuming because of the complex mesh structure. Therefore, the GRU model, which is a simplified model of LSTM, is proposed by choosing a new hidden unit inspired by the LSTM.

The recurrent unit structure of GRU is shown in Fig. 1 and consists mainly of update gates and reset gates, where the importance of time series features is determined by the update gate u_t to capture long-term dependencies, and the memory is reset by the reset gate r_t to remove unnecessary information.

The update gate uses linear calculations and nonlinear processes to define the information to be stored, controls the information transferred from the last hidden state to the current hidden state, and determines the long-term dependency of the time series data. The amount of information stored at the previous moment is proportional to the value of the update gate, which is given by

$$u_t = \sigma(W_u \cdot [x_t, h_{t-1}] + b_u) \tag{1}$$

where, x_t is the input value of GRU at the current moment; h_{t-1} represents the output value at the previous moment; W_u, b_u and σ are the weight, bias and activation function, respectively.

The reset gate determines whether a storage unit's value should be forgotten, helping to understand the short-term dependence of time-series data. The amount of information ignored and forgotten is

$$r_t = \sigma(W_r \cdot [x_t, h_{t-1}] + b_r) \tag{2}$$

where, W_r and b_r represent the weight and bias, respectively. With the reset gate, the new memory content will store past related information using the reset gate with the following equation

$$\widetilde{h}_t = \tanh(W_h \cdot [x_t, r_t \cdot h_{t-1}] + b_n) \tag{3}$$

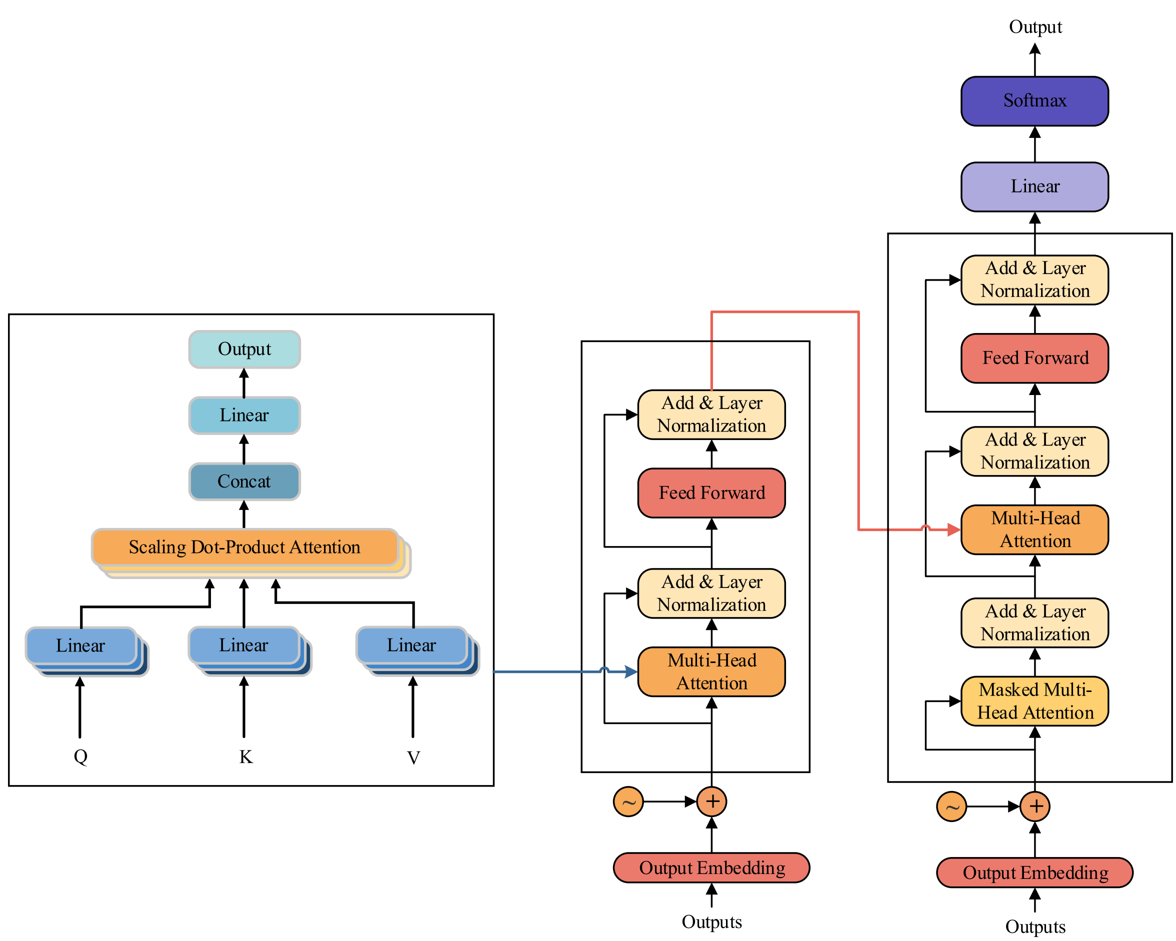

Fig. 3. Structure schematic diagram of Transformer.

Finally, the value of the hidden state is updated such that the information from the current unit will be retained and passed to the next unit. The output of GRU consists of the output information from the previous and the current moment, and the update gate controls the flow of information to and from the two, which is described as

$$h_t = (1 - u_t) \cdot h_{t-1} + \widetilde{h}_t \cdot u_t \tag{4}$$

where, $\widetilde{h}_t$ and h_t denote the previous and current hidden state, respectively.

When processing time series, a unidirectional GRU network can only transmit information in one direction, which means that only information from a point in time in the past is available, while information from a point in time in the future is ignored. To solve this problem, BiGRU deep learning models are built. A BiGRU consists of two layers of GRUs as shown in Fig. 2, which can use both the past and future information.

BiGRU is used to simultaneously extract information from continuous data through interconnected hidden layers. BiGRU performs forward and reverse processing on sequences respectively, ultimately producing the output of the entire network from the forward and reverse outputs, which are described as

$$\begin{cases} \overrightarrow{H} = \left\{ \overrightarrow{h}_1, \overrightarrow{h}_2, \overrightarrow{h}_3, \dots, \overrightarrow{h}_i, \overrightarrow{h}_{i+1}, \dots, \overrightarrow{h}_t \right\} \overleftarrow{H} = \left\{ \overleftarrow{h}_1, \overleftarrow{h}_2, \overleftarrow{h}_3, \dots, \overleftarrow{h}_i, \overleftarrow{h}_{i+1}, \dots, \overleftarrow{h}_t \right\} \end{cases} \tag{5}$$

where, $\overrightarrow{H}$ and $\overleftarrow{H}$ are the forward and reverse hidden state sequence, respectively.

2.2. Transformer

Transformer model is based on an encoder-decoder architecture with input and output embedding, position encoder, multi-head attention module, and feed forward neural network (FFNN). The essence lies in the structure of the encoder and decoder, which consists of modules that can be stacked multiple times. The structure schematic diagram of Transformer is shown in Fig. 3.

1) Input Embedding

Transformer model embeds a vector for the input sequences to determine the position of each sequence and transform the data into the dimensions required by the model through an input embedding process.

2) Position Encoding

Transformer model requires the implementation of a method for dealing with the individual positions of samples, which is handled by assigning a unique value to each sample in the sequence, representing the position of a particular sample in the sequence. As such, the information about the position of each element is stored in the element itself rather than in the model structure.

The positional encoding (PE) module was introduced to obtain the sequential positioning of the input data to understand the problem of modeling continuous time series. The order in which PE is used to store temporal features. PE is calculated using sine and cosine functions based on the position. The order information of the data is encoded by a trigonometric function, and the input vector and PE vector are summed to obtain the position information of the input vector, which is then fed into the encoder as follows:

$$PE(pos, 2i) = \sin\left(pos \ / \ 10000 \frac{2i}{d_{\text{mod } el}} \right) \tag{6}$$

$$PE(pos, 2i+1) = \cos\left(pos \ / \ 10000 \frac{2i}{d_{\text{mod } el}} \right) \tag{7}$$

where, pos denotes the sequence position of the input; $d_{\text{mod } el}$ denotes the input vector's dimension; $i \in \left[0, \frac{d_{\text{mod } el}}{2}\right]$ is the dimension of a single.

3) Encoder

The encoder module is designed to capture the long-term dependencies in the input, and Transformer model can have one or more encoder layers. The encoder structures are all uniform, with each encoder consisting of a linear layer and two identical encoding layers, including a multi-head AM and a fully connected FFNN.

4) Multi-head AM

AM can alleviate the problem of the vanishing gradient problem, which is achieved by providing a direct path to the input vector and helps the neural model to focus on the relevant part of the input sequence when necessary. There are two types of self-attention: scaled dot-product AM and multi-head AM.

The scaled dot product AM sets attention weights to model changes in the time series, which has the advantage of having a strong ability to find dependencies between different positions in the sequence. Considering an input sequence $x = (x_1, x_2, x_3, \dots, x_n) \in \mathbb{R}^{F\times n}$, where each component is characterized by a dimension of F, it is embedded and three new matrices are obtained by x multiplying with three matrices as shown in the following equations

$$Q = xW^Q, K = xW^K, V = xW^V \tag{8}$$

$$Attention(Q, K, V) = Soft \max\left(\frac{QK^T}{\sqrt{d_k}} \right) V \tag{9}$$

where, x is the input matrix; Q, K and V are the query, key and value matrices respectively; $W^Q \in \mathbb{R}^{F\times d_q}$, $W^K \in \mathbb{R}^{F\times d_k}$, $W^V \in \mathbb{R}^{F\times d_v}$ are the matrices to train into the model, d_k is the dimensions of V the vector and $d_q = d_k = d_v = F$. Here, Q, K, and V are all the sensor data of the lithium-ion battery. First, Q and K are dot-processed, and the result of the processing is passed through another linear layer whose output is added to the original matrix after PE, whose the weight coefficients are calculated by $Soft \max$ performing a normalization process, and then the value matrix V is weighted and summed based on the weight coefficients.

In a multi-head attention model, multiple parallel calculations are performed on the attention module, and the individual attention outputs are merged and linearly transformed into the desired dimensions of the former. Intuitively, multi-head attention enables the model to capture different representations of both long-term and short-term dependencies, and allows the model to process different locations while processing the current location. Different location information is learned to synthesize multiple attention heads, resulting in a multi-head attention matrix

$$head_i = Attention\left(QW_i^Q, KW_i^K, VW_i^V \right) \tag{10}$$

$$MultiHead(Q, K, V) = concat(head_1, head_2, \dots, head_k) W^M \tag{11}$$

where, $head_i$ is the ith attention head, W_i^Q, W_i^K, and W_i^V represent the different weight matrix; the $concat$ function is used to stitch the attention heads; W and M are the multi-head attention weight matrix and the number of attention heads, respectively, while W^M is the weight matrix after stitching.

5) Layer Normalization

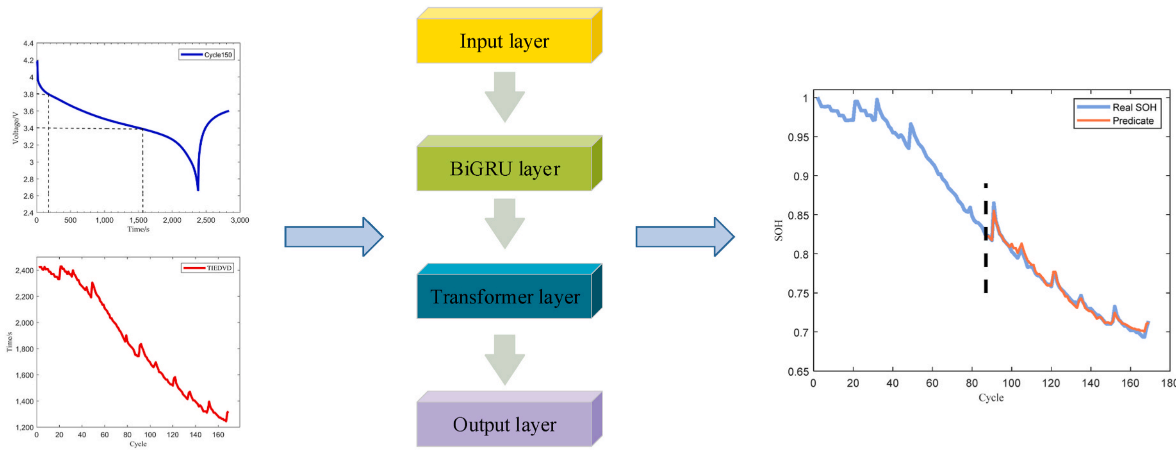

Fig. 4. Structure diagram of BiGRU-Transformer model.

To improve the prediction stability by speeding up the training process and thus obtain the best performance, a residual connection is used followed by performing a layer normalization operation

$$y = Layernorm[Sublayer(x) + x] \tag{12}$$

where, *layernorm* denotes the layer normalization module; *Sublayer* denotes the feedforward neural network module.

6) FFNN

Two linear transformations with ReLU constitute the FFNN, which is represented as

$$FFNN(x) = Linear(\max(0, Linear(x))) \tag{13}$$

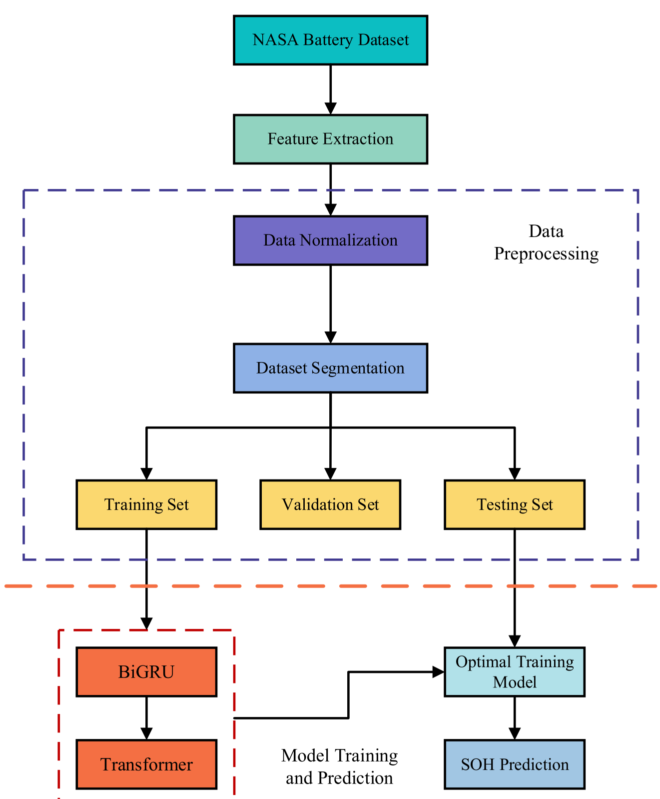

Fig. 5. Prediction flowchart of BiGRU-Transformer model.

The last layer of the encoder is a conventional linear FFNN with ReLU activation functions and discard rates between the hidden layers. In the training process, the backpropagation algorithm updates the weights of each linear layer in each encoder layer, starting with the last FFNN and ending with the linear layer of calculating *Q*, *K*, *V*. The standard configuration has a hidden layer of 2048 units. In the end, the inputs of FFNN is added together as the output of the encoder layer, and layer normalization is carried out.

7) Decoder

The decoder has the same internal structure as the encoder, consisting of two identical layers. In addition, the decoder layer also has an encoder-decoder attention layer, which allows the decoder's component to access the input sequence. The linear layer then converts the size of the decoder output to the size of the matching target sequence and provides the final predicted output. The other adaptive mechanisms of the decoder are the same as those of the encoder. However, the decoder has a second self-correcting mechanism involving the output of the encoder, which is used to compute the new matrices *K* and *Q*, while the matrices obtained from the previous decoder's self-attended head are used to compute the values of the new matrices. Again, the output of the encoder is a memory of the previous value in the sequence, while the matrix generated by the decoder's self-attentive mechanism is the next value in the sequence. The output of the decoder is a calculation of the next value in the sequence, which is used as input to the decoder, and the whole process is repeated until the output sequence is complete.

The Transformer gives an overall overview of the input sequence and finds the real cause of the dependencies. Nevertheless, the absence of CNN and RNN architectures also deprives Transformer models of the ability to capture local features. Thus, we propose a prediction model that combines BiGRU and Transformer in this paper.

3. BiGRU-transformer model

In this paper, BiGRU-Transformer model is proposed, where BiGRU is used to extract time series features and Transformer is utilized to capture long-term dependencies more effectively. We will present the structure of the BiGRU-Transformer model and the SOH prediction process based on BiGRU-Transformer model in this section.

The structure diagram of BiGRU-Transformer model is shown in Fig. 4 and is described as follows.

1) Input layer

In input layer, the indirect HI, which can characterize the capacity during the discharge process of the lithium-ion battery dataset, is extracted and preprocessed. On this basis, the processed dataset is divided into three subsets and the HI of the training set is used as input to the BiGRU layer.

2) BiGRU layer

The BiGRU layer is primarily a BiGRU model, which consists of forward and reverse GRU. The input to BiGRU layer is the HI extracted from the input layer, and the BiGRU model scans the HI sequence forward and backward to capture the sequence features. The sequence obtained after scanning is used as the input of the Transformer layer.

3) Transformer layer

The Transformer layer is a Transformer model. After inputting the sequences obtained from the BiGRU layer, they are then passed into the encoder and decoder of the Transformer so as to study the long-term dependence of capacity degradation in the battery performance data. The correlation between sequence data can be better learned through multi-head AM to improve the accuracy of the prediction model. The results of the Transformer layer learning are transmitted to the output layer for processing.

4) Output layer

The results learned by the Transformer layer are used as the input of this layer. In output layer, the linear layer and $soft\max$ is used to map the predictions in the encoder and decoder, then the SOH predictions are obtained by the best trained model based on the testing set.

From the structure of the BiGRU Transformer model, the flowchart of SOH prediction is shown in Fig. 5, where the SOH prediction process needs four steps.

1) Feature extraction

Considering the complexity and expensiveness of monitoring the internal parameters of batteries such as capacity and internal resistance, in order to predict the SOH of lithium-ion batteries, the first step is to extract the HI that can reflect the capacity degradation based on the charging and discharging voltages, currents, and temperatures curves of lithium-ion batteries. On this basis, the relationship between the capacity and the extracted HI can be obtained through correlation analysis to test the credibility and validity of the extracted HI in characterizing the battery capacity.

2) Data preprocessing

For the extracted HI, the normalization process is performed first and then the processed dataset is divided into three subsets: training set, validation set and testing set. The size of the training set is chosen according to the actual situation. Generally, the larger the training set, the higher the accuracy. Meanwhile, due to the invariance of the overall dataset volume, if the training set is increased, the test set will be reduced accordingly, which will lead to a decrease in the reliability of the test results. Therefore, in the SOH prediction experiments, we select 70 %, 80 % and 90 % of the overall dataset respectively as the training set to attenuate the impact of the size of the training set on the prediction performance of the proposed BiGRU-Transformer model in this paper.

3) Model building and training

The extracted HI serves as the input of BiGRU-Transformer model, which uses BiGRU to learn the hidden states of input features and then passes the learned information as the input to Transformer to learn the long term dependencies, thus completing the BiGRU-Transformer model construction and training. During the training of BiGRU-Transformer model, the mean square error (MSE) is used as the loss function for error evaluation, and the model with lowest training loss is selected as the best trained model.

Table 1
Lithium-ion batteries information in NASA dataset.

Battery	Number of Charges	Number of Discharges	Number of Impedances	Termination Voltage of Discharge
B0005	170	168	278	2.7
B0006	170	168	278	2.5
B0018	134	132	53	2.5

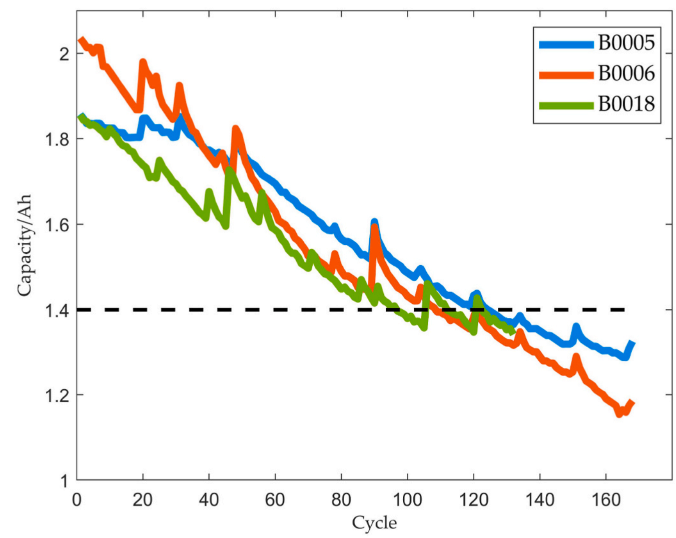

Fig. 6. Capacity degradation curves.

4) Model prediction

In this step, the testing set is input into the obtained best-trained model, and finally the SOH prediction results are obtained. In addition, for the purpose of verifying the effectiveness of the BiGRU-Transformer model, we will evaluate the prediction performance of BiGRU-Transformer model by using several error evaluation metrics in experiments, and analyze the prediction performance by comparing with different models.

Following the flowchart of BiGRU-Transformer model, in the next two sections, we will utilize the BiGRU-Transformer model to predict the SOH of lithium-ion batteries by using the publicly available dataset.

4. Data preprocessing and experimental parameters

To investigate and evaluate the BiGRU-Transformer model, we use the dataset from NASA PCoE. This section is dedicated to introducing the used public dataset, the method of data preprocessing, and giving the model parameter settings.

4.1. Lithium-ion battery ageing dataset

With the aim of training and testing the BiGRU-Transformer model, the batteries B0005, B0006 and B0018 from the public dataset released by NASA PCoE [27] are selected. The dataset from NASA PCoE was collected by the lithium-ion accelerated life test bed. The information for the selected three batteries are given in Table 1. The corresponding capacity degradation curves are shown in Fig. 6, from which the capacities of the three batteries gradually decreased over time, with the recovery of capacities accompanying the degradation process, and the

Table 2
Dataset splitting.

Battery	Training Set	Validation Set	Testing Set
B0005	(70 %)50	37	(30 %)37
	(80 %)74	25	(20 %)25
	(90 %)99	12	(10 %)13
B0006	(70 %)43	32	(30 %)33
	(80 %)65	22	(20 %)21
	(90 %)86	11	(10 %)11
B0018	(70 %)38	29	(30 %)29
	(80 %)58	19	(20 %)19
	(90 %)77	10	(10 %)9

Table 3
Parameter setting.

Parameter	Setting Value
Optimizer	Adam
Loss Function	MSE
Activation Function	ReLU
Batch Size	32
Number of Iterations	260
Discard Rate	0.2
Learning Rate	0.005
Number of Multi-Head Attention Heads	8
Dimensionality of Model	16
Number of Hidden Layers	1
Number of Hidden Layer Neurons	16

end of life (EOL) is reached when the nominal capacities are reduced by 30 %.

4.2. Data preprocessing

A noticeable indication of battery degradation is a decrease in its capacity. Therefore, for lithium-ion batteries, we use the capacity to define the SOH as

$$SOH = \frac{C_{actual}}{C_{nom}} \tag{14}$$

where, C_{actual} and C_{nom} represent the actual and rated capacity, respectively.

In order to avoid anomalous data and to take into account the effects of data magnitude, the data are cleaned and regularized first. The data undergo a cleaning process to eliminate outliers, and any missing values are estimated using techniques such as moving averages or interpolation with intermediate values.

The data preprocessing ensures that the confounding effects of erroneous model values are avoided and the periodic averaging is performed to avoid the effects of short-term fluctuations. In ML algorithms, data normalization is frequently employed to mitigate the negative impact of varying value ranges. It helps enhance the speed and accuracy of model convergence. In this paper, min-max normalization, where the data are scaled between 0 and 1, is used, i.e.,

$$x_n = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{15}$$

where, x_n and x denotes the processed data and original data, respectively, while $x_{\max}$ and $x_{\min}$ correspond to the maximum and minimum values of the original data, respectively.

4.3. Model parameter settings

In the experiments of this paper, the splitting of the datasets for the three selected batteries are shown in Table 2. These datasets were divided into three subsets: training set, validation set, and testing set. The training set was utilized for model training, the validation set for parameter tuning, and the testing set for evaluating the accuracy of the model.

On the other hand, the appropriate parameters help to improve the predictive accuracy of the BiGRU-Transformer model. For the selection of model parameters, grid search methods and cross validation methods are usually used to obtain the optimal parameters. However, the K value in cross validation may affect the sensitivity of training set changes, thereby affecting the results of hyperparameters. Therefore, we use the grid search methods to determine hyperparameters in this paper. Table 3 gives the parameters in the experiments of this paper.

Optimizer: The parameters are optimized by using Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.98$, and the learning rate is derived by

$$lrate = (init\_lr \cdot warmup\_steps)^{0.5} \cdot \min \cdot \left(step\_num^{-0.5}, step\_num \cdot warm\_steps^{-1.5}\right) \tag{16}$$

where, $warm\_steps = 4000$.

Regularization: Discard rate regularization allows the methods for training different neural network architectures to run in parallel. Discard rate refers to a random number of nodes, both hidden and visible nodes. However, these nodes are ignored, forcing layer nodes to take probabilistic responsibility for their inputs, as the discard rate adds noise to the training process. The discard rate factor for each encoder and decoder sublayer is 0.2.

5. Experiment results and comparative analyses

5.1. Performance evaluation metrics

In this section, three error evaluation metrics, i.e., root mean square error (RMSE), mean absolute error (MAE) and mean absolute percentage error (MAPE), are used to quantitatively assess the accuracy of the proposed SOH prediction model, which are defined as

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(z_i - \hat{z}_i\right)^2} \tag{17}$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|z_i - \hat{z}_i| \tag{18}$$

$$MAPE = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{z_i - \hat{z}_i}{z_i}\right| \times 100\% \tag{19}$$

where, z_i denotes the real SOH, and $\hat{z}_i$ denotes the predicted SOH.

5.2. HI extraction and correlation analysis

HI extraction is essential to characterize the degradation process. In general, the degradation process of a battery is usually described by the capacity and internal resistance [28]. Unfortunately, the meters of monitoring capacities and internal resistance are very expensive, and it takes a long time to monitor online. And there are many complex factors, such as materials and chemical reactions, which severely limit the monitoring of these internal parameters in practical applications and prevent online acquisition and real-time monitoring [29]. For these reasons, in this paper, easily measurable and more cost-effective voltage and temperature measurement instruments, which are used to extract HI, are chosen to solve the problem of difficult access and online monitoring of SOH in lithium-ion batteries.

The SOH of lithium-ion batteries can be reflected by the capacity, and the HI extracted during charging and discharging is a characteristic parameter with a strong correlation with the capacity, so the extracted indirect HI is a reliable reflection of the SOH [30]. Therefore, we use indirect HI extracted from the voltage and temperature [31–33] to predict SOH in this paper. We take the battery B0005 as an example,

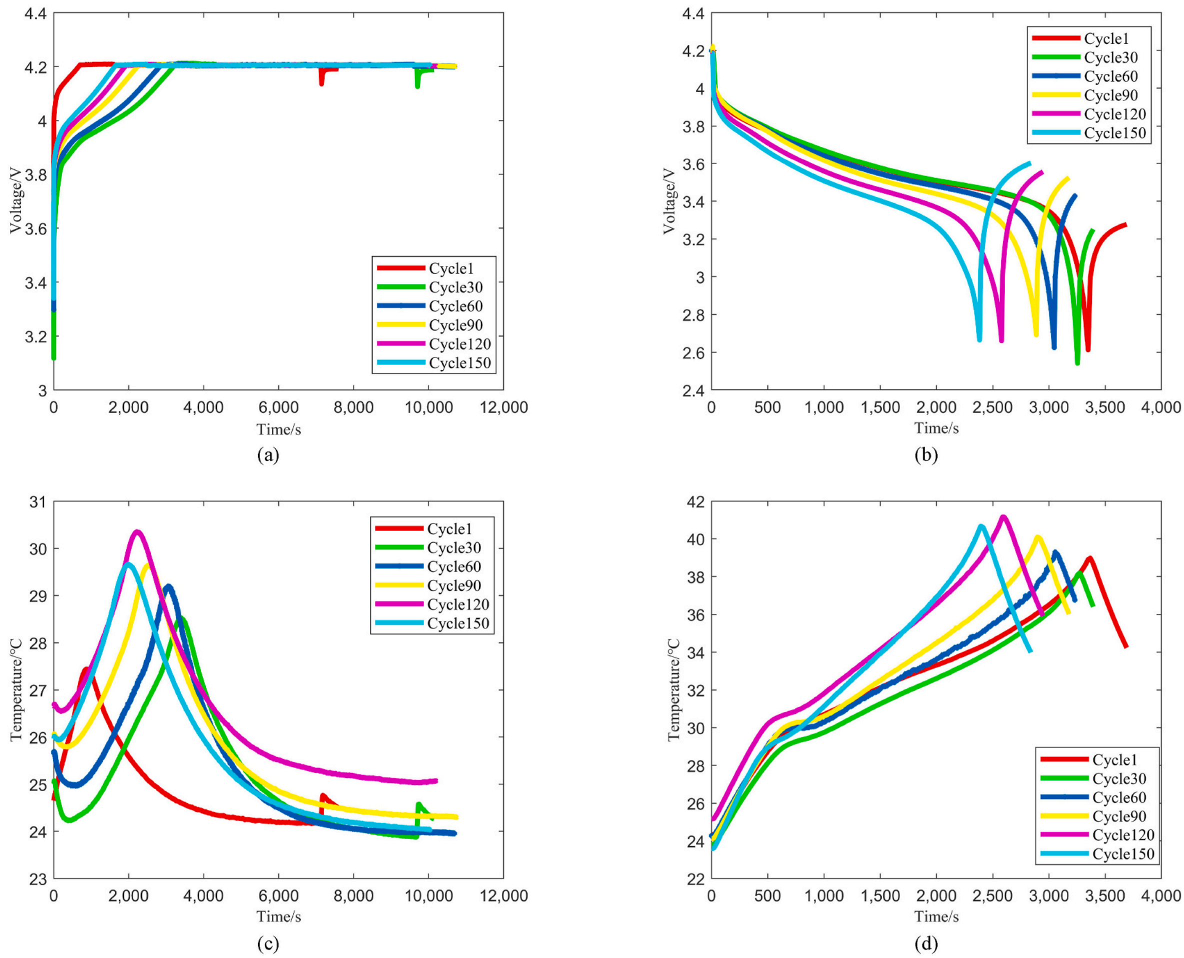

Fig. 7. Voltage and temperature curves of battery B0005 at different cycles. (a) Charge voltage curves; (b) Discharge voltage curves; (c) Charge temperature curves; (d) Discharge temperature curves.

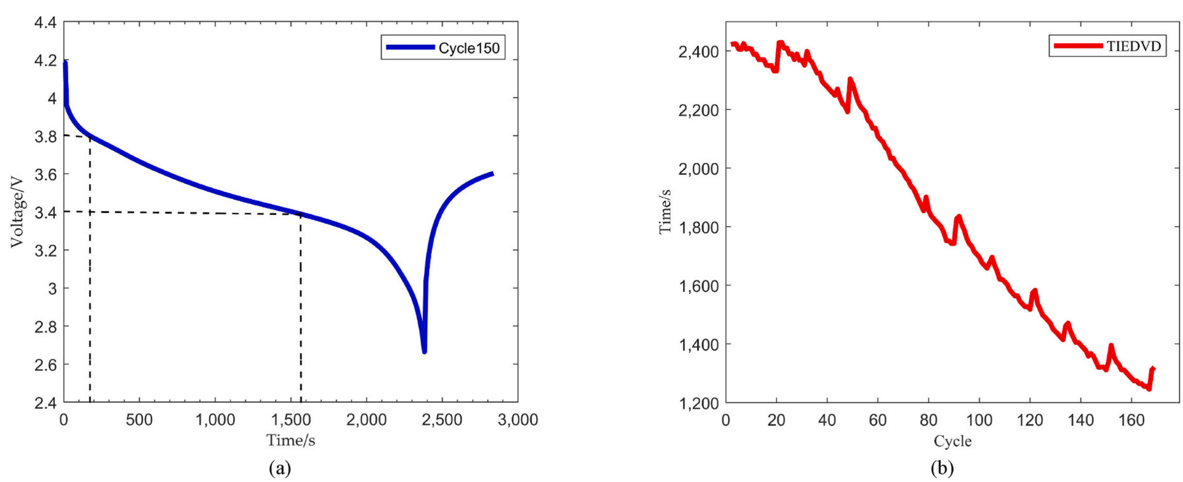

Fig. 8. Extraction of HI. (a) Discharge voltage curve at the 150th cycle; (b) TIEDVD.

which is shown in Fig. 7. The charging process in Fig. 7(a) results in a gradual increase in voltage slope and a decrease in charging time. Because the voltage variation range is small within a certain time interval, the discharge process is analyzed in this paper. Fig. 7(b) illustrates the voltage variation of battery B0005 during the discharge process at different cycles. As the battery cycle increases, the rate of voltage drop gradually accelerates, i.e., the time required to discharge becomes shorter and shorter. Therefore, it makes sense to extract HI using the hidden signs of degradation shown by the discharge voltage.

We still take the battery B0005 as an example to extract HI. Fig. 8(a) shows the 150th cycle of battery discharge voltage. The discharging process of the battery can be observed in Fig. 8(a), which is predominantly divided into three stages. The first stage is characterized by a rapid decline in voltage, from the initial voltage to approximately 4 V.

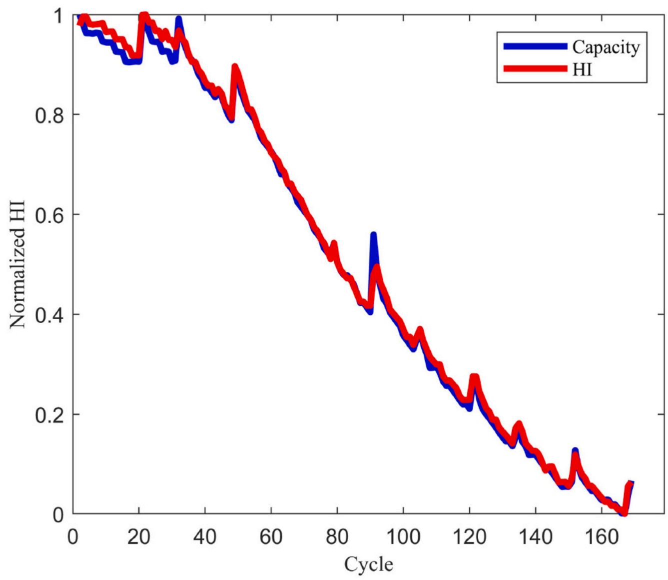

Fig. 9. Normalized capacity and HI degradation curves for battery B0005.

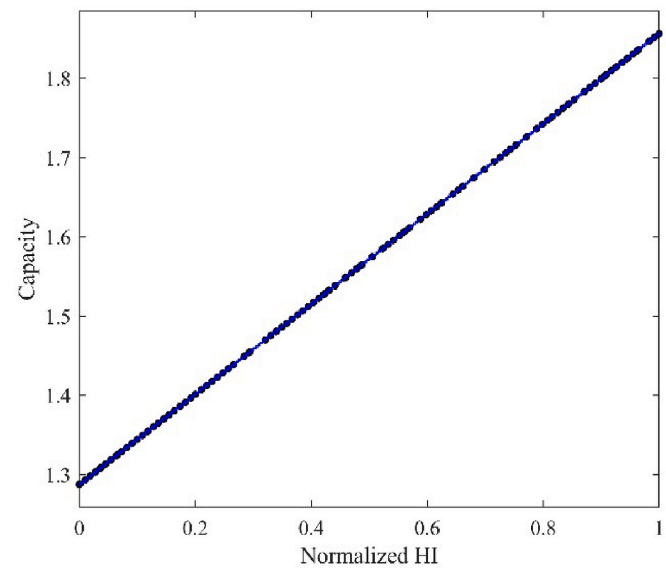

Fig. 10. Scatter plots of normalized HI and capacity.

Table 4
Correlation analyses between HI and capacity.

Correlation analysis	B0005	B0006	B0018
Pearson	0.9972	0.9969	0.9973
Spearman	0.9983	0.9995	0.9986

The second stage is a plateau discharge phase, with a stable voltage drop from around 4 V–3.4 V. Finally, the third stage is marked by a rapid voltage drop, leading to the battery reaching the cut-off voltage. Fig. 8 (a) shows that the voltage between 3.4 V and 3.8 V is rich in degradation information, while Fig. 8 (b) shows the extracted time interval of an equal discharging voltage difference (TIEDVD), where the time interval corresponding to TIEDVD in *i*th cycle is

$$TIEDVD = |t_{V_{MAX}} - t_{V_{MIN}}| \tag{20}$$

where, $t_{V_{MAX}}$ and $t_{V_{MIN}}$ are the time corresponding to the high and low voltage value, respectively. Ultimately, the proposed HI can be expressed as

$$TIEDVD = \{TIEDVD_1, \ldots, TIEDVD_i, \ldots, TIEDVD_n\} \tag{21}$$

Fig. 9 shows the TIEDVD extracted from the battery B0005, from which the proposed HI can effectively reflect the degradation process, and the HI curve presents a monotonic trend consistent with the capacity curve.

To select valid HI, feature selection is necessary to reduce data preprocessing time and improve the evaluation efficiency of ML algorithms. Correlation coefficient methods are widely used for HI selection. The curves of HI and capacity have the same monotonic trend, so it is difficult to determine the correlation between HI and capacity directly, which is addressed by the Pearson correlation coefficient here. When the absolute value of the Pearson correlation coefficient is exactly 1, it means that all data points lie on the same line in a linear equation, while a correlation coefficient equal to 0 means that there is no linear relationship between the variables. If the coefficient is positive, it indicates a linearly positive correlation between the variables. Conversely, if it is a negative integer less than zero, then the relationship is inversely proportional and negatively correlated. Unlike Pearson correlation, which estimates linear relationships, Spearman correlation estimates a monotonic (non-linear) relationship. If both Pearson and Spearman correlation analyses are approximately equal to 1, it means that the two variables are monotonic and linear.

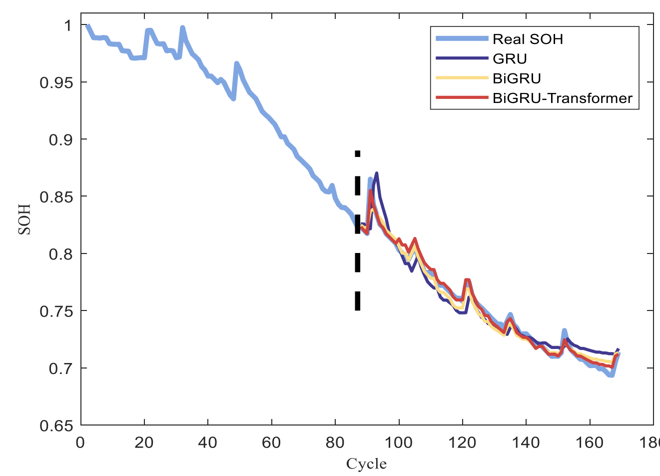

Fig. 11. SOH prediction for battery B0005 by using GRU model, BiGRU model and BiGRU-Transformer model at SP = 70 %.

The correlation analyses between HI and capacity are shown in Table 4, from which we can see that the correlation coefficients for the three batteries are all close to 1 and above 0.99, indicating a significant linear correlation between HI and capacity. To further illustrate that HI can represent the capacity to characterize the degradation, we continue with the qualitative analysis. A scatter plot of the normalized HI versus capacity is given in Fig. 10 to show the correlation, where dots represent HI and straight lines represent capacity. From Fig. 10, HI has a strong linear correlation with the capacity, and HI decreases with the capacity, indicating that HI is positively correlated with the capacity. Therefore, both the quantitative and qualitative analyses demonstrate that the extracted HI fully characterizes the degradation process, based on which we build a deep learning model for SOH prediction such that the expensive instrumentation and complex operations can be avoided.

5.3. SOH prediction results and comparative analysis

In this subsection, the proposed BiGRU-Transformer model is utilized to predict the SOH at different start points (SPs). To preliminarily verify

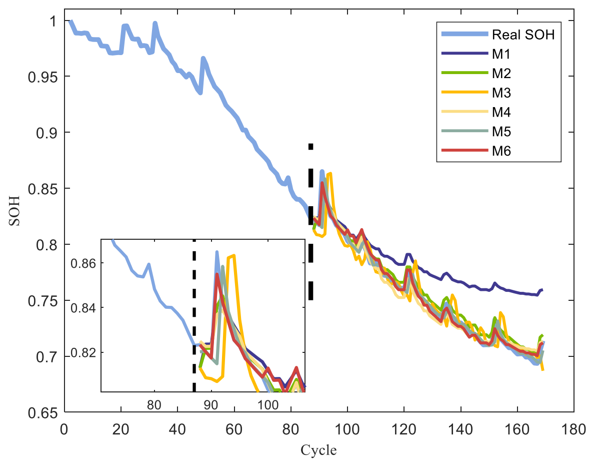

Fig. 12. SOH prediction for battery B0005 by using different models at SP = 70 %.

the effectiveness of BiGRU-Transformer model, two related models, i.e., GRU model and BiGRU model, are designed for the comparison experiments first. The battery B0005 is selected as an example for SOH prediction at SP = 70 %, i.e., the cycle that corresponds to 70 % of the cycle number corresponding to the EOL is set to the SP. The prediction results of battery B0005 are presented in Fig. 11, from which the prediction trends of all three models are roughly close to each other, indicating the advantage of NNs in SOH prediction. In Fig. 11, the GRU model shows the largest error, followed by the BiGRU model, which is able to learn the hidden state of the input features compared with GRU model to reduce the prediction error. The BiGRU-Transformer model shows the best prediction performance in both capacity degradation and capacity generation, and the prediction curves are the closest to the real values, which verifies the effectiveness of the BiGRU-Transformer model.

Moreover, the BiGRU-Transformer model is also compared with RVM [34], SVR [15], CNN, BiGRU and CNN-BiLSTM-AM models to showcase its advantage. For the convenience of description, RVM, SVR, CNN, BiGRU, CNN-BiLSTM-AM and BiGRU-Transformer models are denoted as M1, M2, M3, M4, M5 and M6, respectively. We take battery B0005 as an example and SP = 70 %. In this setting, the SOH prediction results for battery B0005 by using different prediction models are shown in Fig. 12, from which the prediction trends of the six models are broadly similar, indicating the strength of NNs in SOH prediction. In Fig. 12, M1 model has the highest error, followed by M3 model. In M5 model, BiLSTM is employed to extract time series information from both directions. However, its predictions for capacity recovery still fall slightly behind that of M6 model. The M6 model shows the best prediction performance in trend decay and capacity recovery, with the prediction curves closest to the real values.

To further demonstrate the prediction accuracy of the BiGRU-Transformer model (M6 model), the SOH prediction for battery B0005 at different SPs are presented in Fig. 13, from which the prediction trends of models M1 and M2 are closer to each other at SP = 80 %, while the M6 model is closer to the actual value and also shows better prediction results. The reason is that the BiGRU of the M6 model learns the hidden states of the input characteristics, and then passes them to the encoder and decoder to investigate the long-term dependence of the battery performance data on capacity decline, which further highlights the correlation between the serial data. On the other hand, from Fig. 13, battery B0005 also achieves more accurate prediction results at SP = 90 % than that at SP = 80 %, particularly in terms of capacity recovery.

In this paper, RMSE, MAE and MAPE, which have been described and defined in subsection 5.1, are used as the error indicators. Table 5 presents the SOH prediction results for batteries B0005, B0006, and B0018 using various models at different SPs. As shown in Table 5, for the error evaluation metrics RMSE and MAPE, the M6 model is better than M1, M2, M3, M4, and M5, while the M6 model is better than M1, M2, M3, and M4 for the error evaluation metric MAE, which means that the M6 model has better prediction accuracy. On the other hand, for a same battery, such as battery B0005, when the training data is increased, i.e., when the SP is increased from 70 % to 80 %, the RMSE and MAPE of the M6 model decrease from 0.00729 to 0.343 to 0.00243 and 0.191, respectively, which is consistent with natural cognition and normal logic. Besides, for the M6 model, the MAEs are only slightly inferior to those for the M5 model at SP = 70 % for the battery B0005, at SP = 80 % for the battery B0018 and at three SPs for the battery B0006. The reason is that the huge fluctuation near the capacity decline curve (e.g., the battery B0005 at the 90th cycle to the 98th cycle) will affect the learning effect of M6 model, which means that there is still room to enhance and improve M6 model. However, in the validation experiment, it is found that when the SP is set before the 90th cycle of battery aging, that is, before the huge fluctuation point, the prediction performance of the M6 model is still better than that of the M5 model. We can also formulate a strategy based on the above results, if the SP is near the inflection point of the capacity decline curve, we can try to use the M5 model, and in other cases, we still use the M6 model for SOH prediction.

According to the experiment results, it can be concluded that the factors that affect the SOH prediction accuracy of the model proposed in this paper are as follows.

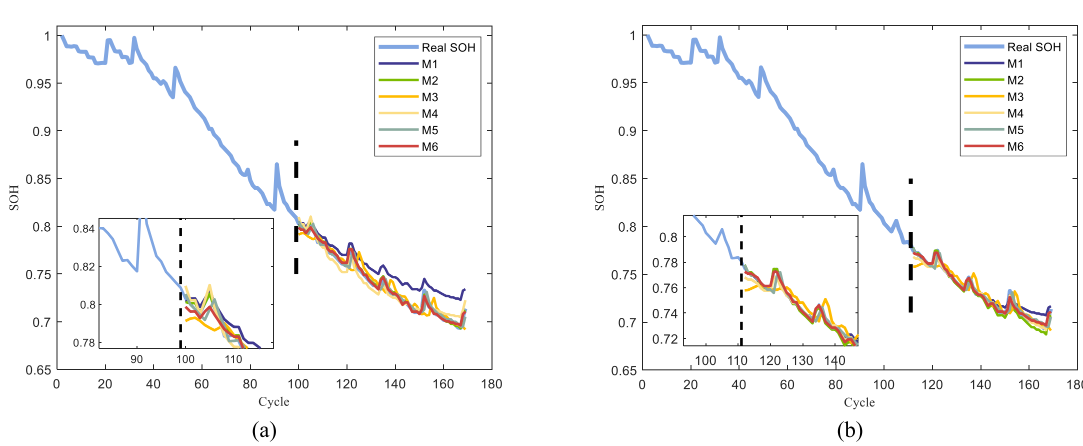

Fig. 13. SOH prediction results for battery B0005 based on different models. (a) SP = 80 %; (b) SP = 90 %.

Table 5
SOH prediction results by using various models at different SPs.

Battery	Prediction SP	Model	RMSE	MAE	MAPE (%)
B0005	87 (70 %)	M1	0.0238	0.0207	2.735
		M2	0.00836	0.00698	0.892
		M3	0.00893	0.00565	0.738
		M4	0.00781	0.00564	0.740
		M5	0.00737	**0.00382**	0.496
		M6	**0.00729**	0.00432	**0.343**
	99 (80 %)	M1	0.00944	0.00649	0.893
		M2	0.00571	0.00427	0.505
		M3	0.00863	0.00661	0.886
		M4	0.00781	0.00564	0.740
		M5	0.00481	0.00316	0.425
		M6	**0.00243**	**0.00261**	**0.191**
	111 (90 %)	M1	0.00746	0.00712	0.859
		M2	0.00683	0.00709	0.842
		M3	0.00813	0.00633	0.825
		M4	0.00487	0.00452	0.479
		M5	0.00487	0.00307	0.420
		M6	**0.00223**	**0.00233**	**0.172**
B0006	75 (70 %)	M1	0.0677	0.0531	0.747
		M2	0.0277	0.0205	0.738
		M3	0.0151	0.0101	1.49
		M4	0.0150	0.0108	1.63
		M5	0.0114	**0.00485**	0.709
		M6	**0.00823**	0.00997	**0.650**
	87 (80 %)	M1	0.0658	0.0506	0.909
		M2	0.0241	0.00551	0.807
		M3	0.0134	0.00864	1.28
		M4	0.0133	0.00836	1.24
		M5	0.0105	**0.00491**	0.727
		M6	**0.00832**	0.00906	**0.602**
	97 (90 %)	M1	0.0182	0.0138	1.40
		M2	0.0117	0.00955	1.15
		M3	0.00914	0.00708	1.11
		M4	0.00870	0.00705	1.10
		M5	0.00572	**0.00361**	0.561
		M6	**0.00471**	0.00611	**0.386**
B0018	67 (70 %)	M1	0.0805	0.0685	2.14
		M2	0.0317	0.0279	1.35
		M3	0.0145	0.0110	1.44
		M4	0.0114	0.00710	0.917
		M5	0.0108	0.00621	0.797
		M6	**0.00393**	**0.00423**	**0.323**
	77 (80 %)	M1	0.0197	0.0158	2.13
		M2	0.0157	0.0136	1.77
		M3	0.0153	0.0121	1.60
		M4	0.0112	0.00679	0.890
		M5	0.0109	**0.00623**	0.619
		M6	**0.00651**	0.00687	**0.519**
	87 (90 %)	M1	0.0186	0.0168	1.92
		M2	0.0174	0.0144	1.89
		M3	0.0153	0.0121	1.60
		M4	0.0134	0.00943	1.24
		M5	0.0112	0.00627	0.828
		M6	**0.00394**	**0.00378**	**0.286**

(1) Model structure and parameters: We combine the advantages of BiGRU and Transformer in this paper. Compared to LSTM units, GRU units have higher computational efficiency and fewer parameters such that BiGRU can extract local features of time series, while Transformer has excellent global information capture ability. The model parameters, including the number of neurons, the number of hidden layers, and the number of multi-head attention heads, have an impact on prediction accuracy. By applying optimization algorithms, the better model parameters can be found to improve prediction accuracy.
(2) Selection of SP: From Figs. 12 and 13, it can be seen that the later the SP is, the more training data is used, and the better the prediction accuracy of the model.
(3) Huge fluctuations caused by capacity recovery: From Fig. 12 and Table 5, there are significant fluctuations during the 90th to 95th cycles of battery aging, which have an impact on the MAE indicators of the model. To solve this problem, it is possible to choose an earlier SP or adopt different model prediction strategies, that is, choose a suitable model for prediction based on whether there are significant fluctuations.

6. Conclusions

In this paper, a hybrid model by combining BiGRU and Transformer was developed to predict the SOH of lithium-ion batteries. The proposed BiGRU-Transformer model utilizes the structures and advantages of BiGRU and Transformer such that it can predict the SOH with high accuracy by learning the hidden states of input features and the long term dependency relationship. The SOH prediction results based on the public dataset present the advantages of BiGRU-Transformer model in accuracy, robustness and generalisation capability. The further works can be considered as follows: (1) SOH prediction of lithium-ion batteries based on the actual usage scenarios, such as studying SOH prediction of lithium-ion batteries for unmanned aerial vehicles, electric vehicles, etc. (2) The hybrid prediction method based on the physicochemical laws of lithium-ion batteries and artificial intelligence methods.

CRediT authorship contribution statement

Chenyu Jia: Data curation, Methodology, Writing – original draft. **Yukai Tian:** Methodology, Formal analysis, Writing – review & editing. **Yuanhao Shi:** Investigation, Validation. **Jianfang Jia:** Software, Formal analysis, Funding acquisition. **Jie Wen:** Supervision, Methodology, Writing – original draft. **Jianchao Zeng:** Supervision, Conceptualization, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in this paper comes from public datasets.

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